Pipeline to make SO database (w/ current problems)

① Whitelist input →

Register of Antarctic Marine Species
↳ DarwinCore Database files

filter
+ Genbank identifier
+ species/subspecies
+ ~~taxonomic level~~
+ ~~filter~~ locality data

Raw .csv of species + localities

☆ Bash/rosalind get all lineages for dawinCore taxonKIT

filter by locality in Darwin core
+ Cross reference w/ Ocean Biodiversity (OBIS) Information System

.R

① csv automatically approved

② csv "flagged" taxa for manual curation

manual curation #2

merge complete whitelist

.R visualise SO groups/ taxonomic distribution

↳ create target groups

Problem/Road Bump:
Where r all the Algae??? ∵ new problem
(have a fix)

this could be easiet point to fix these problems

Download all genomic (nuclear + organel) Genbank + RefSeq records
↳ get lineages as well

visualise + stringent filter non-SO genomes

.Rmd, visualise and filter non SO, goal: Big(ish) DB; focus on SO taxa

- deduplicate records
  - ↳ per organelle, 1 record per taxa
  - ↳ only keep complete organelles
  - ↳ coverge >25
  - ↳ gap length ratio 1.>
  - ↳ only keep ① chromosome ② complete genome assemblies
  - ↳ filter by NSO/LSO

Problem/Road Bump:
very large DB that has large representation from non-target groups. including insects + flowering plants

.csv w/ all metadata non SO hits

filter non-SO by target groups made earlier

filter out all southern Ocean taxa (genomes)
↳ these stay in no curation

visualise amounts, actually quite small

/create

merge lists! in Rmd.

expand lineages
visualise DB dist

- subtract all APES

taxonomic grouped master accession list

bash/rosalind ftp all accesion files

raw DB on rosalind in .fna

split dB files into chunks

Benchmark resources

index parts in parallel